BEHIND THE SCREEN

SUBTOPIC 2

Inside the Cloud: Data Centres and Physical Infrastructure

A detailed research report for an educational website

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Central question
What physical systems store, process, and move digital information—and how do they keep online services available?

Purpose: explain the physical reality behind cloud computing in clear language, while distinguishing general technical explanations from documented examples and claims that require site-specific evidence.

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This report follows the Subtopic 2 requirements in the HM216 Group Briefing Report: a beginner-friendly explanation, labelled component diagram, documented example, sources, and a proposed interactive layout.

1. Executive summary

The word “cloud” can make digital services sound weightless or distant from the physical world. In practice, cloud services depend on buildings filled with computing equipment, storage devices, network connections, electrical systems, cooling equipment, security controls, and staff. A data centre is a facility designed to house and operate this equipment. Some facilities support a single organisation; others provide computing and storage services to many customers.

When a user opens a website or an application requests information, the request may reach servers in one or more data centres. Software running on those servers processes the request, reads or writes data, and sends a response through network equipment. The exact route depends on the service design, location, caching, load balancing, and current network conditions. There is no single route that applies to every click.

A data centre must do more than run computers. It must deliver stable power, remove heat, protect equipment, connect to networks, monitor faults, and recover from failures. Redundancy—having alternate equipment or paths—can improve reliability, but it increases complexity and cost and does not guarantee that a service will never fail.

The central finding is that the cloud is a service model built on physical infrastructure. Understanding that infrastructure helps explain both the convenience of digital services and their dependence on electricity, engineering, maintenance, security, and careful design.

> *Scope note: This report explains common principles. Individual facilities differ, and any precise statement about a particular company, data centre, outage, energy use, or reliability target should be checked against a dated, direct source.*

2. What is a data centre?

2.1 A practical definition

A data centre is a building or dedicated facility that houses information technology equipment and the supporting systems needed to operate it. The IT equipment commonly includes servers, storage devices, and network equipment. Supporting infrastructure includes power distribution, backup power, cooling, fire detection and suppression, physical security, and environmental monitoring.

The facility is engineered to keep equipment within operating conditions and to provide connectivity to other systems. Depending on its purpose, it may host public cloud services, business applications, databases, websites, scientific workloads, telecommunications services, or internal systems.

2.2 Data centre, server room, and cloud provider

A small organisation may keep a few servers in a locked room. A larger data centre has purpose-built systems for power, cooling, connectivity, and maintenance. A cloud provider operates computing resources and offers them as services over a network. These concepts overlap but are not identical: a cloud service may run across several facilities, and a data centre can host systems that are not offered as public cloud services.

2.3 Why buildings matter

The building is part of the technical system. Floors, cable pathways, access controls, fire protection, cooling distribution, and power routes affect how equipment is installed and maintained. A facility can contain sophisticated servers yet still be vulnerable if its power, cooling, network connections, or operating procedures are poorly designed.

USER DEVICE → INTERNET / PRIVATE NETWORK → DATA-CENTRE NETWORK → COMPUTE + STORAGE → RESPONSE RETURNS

This is a simplified conceptual path, not a trace of any particular visitor's real traffic.

3. Cloud computing: a service, not a place in the sky

3.1 What “cloud” means

Cloud computing describes providing computing resources—such as processing, storage, databases, and software—through a network. Instead of buying and operating every server directly, a customer can request resources from a provider. The provider manages some or all of the underlying infrastructure, depending on the service model.

The word cloud describes how a user accesses a service, not the absence of hardware. The hardware still exists somewhere, uses electricity, needs maintenance, and connects to networks.

3.2 Shared infrastructure and virtualisation

Virtualisation allows software to present a physical computer's resources as multiple logical machines or environments. Containers provide another way to package applications and their dependencies. These technologies can help providers allocate resources flexibly, isolate workloads, and use hardware more efficiently. They do not eliminate the physical limits of processors, memory, storage, cooling, or network capacity.

3.3 Why users may not know the location

A cloud service may use multiple facilities or regions. A provider can place workloads near users, distribute copies of data, or move processing as demand changes. The user usually sees a service address or application interface rather than a map of the servers. For this reason, an educational diagram should explain possible arrangements rather than claim to show the actual location of a visitor's data.

3.4 Service models in simple terms

• **Infrastructure as a Service (IaaS):** customers rent virtual computing, storage, and networking resources and manage much of the software stack.

• **Platform as a Service (PaaS):** customers deploy applications using a managed platform, while the provider manages more of the underlying environment.

• **Software as a Service (SaaS):** users access a complete application; the provider operates most of the technical stack.

These categories describe responsibility boundaries, not different laws of physics. All depend on physical equipment somewhere.

4. The main physical components

4.1 Servers

A server is a computer that provides a service to other computers. It may run a web server, application code, database software, authentication, search, media processing, or other workloads. A server contains processors, memory, local storage or storage connections, network interfaces, and power supplies. Different workloads require different combinations of CPU, memory, accelerators, and input/output capacity.

4.2 Storage systems

Storage systems keep information when it is not actively being processed. They may use solid-state drives, hard disk drives, storage arrays, or distributed systems made from many devices. Storage software can manage replication, error detection, access permissions, and recovery. A single drive is not the same as a complete storage service: reliability also depends on software, configuration, copies, and operational practice.

4.3 Network switches and routers

A network switch connects devices within a local network, forwarding frames based on link-layer information. A router forwards packets between different IP networks. Data centres commonly use layered network designs, with multiple switches and routers connecting servers to other racks, buildings, providers, and the wider internet. The precise design varies by facility and service.

4.4 Racks, cabling, and fibre

Servers and network devices are often mounted in racks. Copper cabling may connect equipment over short distances, while optical fibre carries data as pulses of light and supports high-capacity links. Cable management is important because poor airflow, accidental disconnection, or difficult-to-trace cables can complicate maintenance.

4.5 Supporting systems

• Power distribution units deliver electricity to equipment in a controlled way.

• Uninterruptible power supplies (UPS) can bridge short interruptions and provide time for transition to another source.

• Generators or other backup sources can supply power during longer grid outages, subject to fuel, start-up, maintenance, and operating constraints.

• Cooling equipment moves heat away from IT hardware.

• Sensors and monitoring systems track temperature, humidity, power, network status, and equipment health.

RACK: [SERVER] [SERVER] [STORAGE] [SWITCH] | POWER + NETWORK + AIRFLOW

5. How a request is processed

5.1 A simple example

Imagine a student opens a course website. The browser sends a request through the student's device and local network. Internet routing carries it toward the service. The provider may use a front-end system or load balancer to select a healthy server. Application software then decides what information is needed, perhaps querying a database or retrieving a stored file. The response is sent back through the network to the browser.

5.2 Front end, application, and database

A web service may divide work among several layers. The front end accepts connections and serves static content or passes requests onward. Application servers run business logic, check permissions, and decide what to do. Database systems organise structured records, while object or file storage may hold documents, images, and videos. Some systems combine these roles; others separate them for scale and security.

5.3 Load balancing

A load balancer distributes requests among available servers according to rules and health checks. It can help prevent one server from receiving too much work and may stop sending traffic to an unhealthy instance. It cannot solve every problem: a shared database, network bottleneck, faulty software release, or regional outage can still affect many servers at once.

5.4 Caching

A cache keeps a copy of frequently requested data closer to where it is needed. A browser, content delivery network, proxy, or application may serve a cached result instead of retrieving the original every time. Caching can reduce latency and demand on origin systems, but cached data must be managed carefully when information changes or access rights differ.

5.5 Why there is no universal route

The path depends on DNS results, network routing, service architecture, geographic location, load, caching, and failover decisions. Two requests to the same website can be handled differently. A diagram should therefore be labelled “typical simplified journey” rather than presented as a universal map.

6. How data is stored and retrieved

6.1 Bits, files, and databases

Digital information is represented as bits. Applications interpret those bits as text, images, video, records, or other formats. A file system organises files and directories, while a database provides structured methods for storing and querying records. At the physical level, storage devices encode data in electronic or magnetic states; software provides the higher-level organisation users recognise.

6.2 Replication and backups are different

Replication maintains copies of data across devices or systems, often to support availability or performance. A backup is a separate recovery copy designed to restore data after accidental deletion, corruption, attack, or other loss. Replication alone may copy a mistake or malicious change to every replica, so it is not a full replacement for well-designed backups.

6.3 Durability and consistency

Durability concerns whether acknowledged data remains available after failures. Consistency describes what users see when data changes, and different systems offer different guarantees. Some systems prioritise immediate agreement between copies; others allow temporary differences to improve availability or performance. These are design choices with consequences for applications.

6.4 Encryption and access control

Encryption can protect data while it travels over a network and while it is stored. Access controls determine which users or services may read or change information. Security depends on correct key management, identity systems, software updates, logging, and operational practice—not encryption alone.

6.5 A practical example

When a user uploads an assignment, the application may authenticate the user, check permissions, transfer the file, store it in an object-storage system, record metadata in a database, and return confirmation. The service may create additional copies or process the file asynchronously. The exact implementation is provider-specific, so the example should be presented as one plausible architecture.

7. Network equipment and connectivity

7.1 Inside the facility

Servers communicate through network interfaces and switches. Links may be grouped to provide capacity or resilience. Network design must consider bandwidth, latency, congestion, segmentation, and the consequences of a failed device or cable. Redundant links help only if they do not share a hidden common point of failure.

7.2 Connecting outside

Data centres connect to telecommunications providers and other networks through fibre links and routing equipment. Large facilities may connect to multiple carriers and physically separate routes. Diverse routes can reduce the risk that one cable cut interrupts all connectivity, but route diversity must be real rather than merely contractual.

7.3 Latency and bandwidth

Latency is the time taken for data to travel and be processed. Bandwidth is the capacity of a link to carry data over time. A connection can have high bandwidth but still feel slow if latency is high or servers are overloaded. Congestion, distance, packet loss, and application design can all affect the user experience.

7.4 Content delivery networks

A content delivery network (CDN) distributes copies of eligible content across locations closer to users. A nearby cache can serve images, scripts, or video without every request travelling to the origin facility. Dynamic or personalised requests may still need to reach application systems. A CDN changes the likely path but does not make the infrastructure disappear.

7.5 Network failure example

A fibre cut can interrupt a link, but the impact depends on route diversity, routing protocols, spare capacity, and how quickly operators respond. If alternate paths exist and are configured correctly, traffic may reroute with limited disruption. If several routes share the same physical corridor, a single incident can affect them together.

8. Electricity, backup power, and resilience

8.1 Why power quality matters

Servers need a stable supply of electricity. Voltage disturbances, outages, or power-distribution faults can cause equipment to stop or restart. Facilities design power distribution to meet the load, isolate faults, support maintenance, and reduce the chance that a single failure affects the entire site.

8.2 UPS and generators

A UPS typically provides short-term power continuity and may condition power or protect against certain disturbances. Generators can provide longer-duration backup when the grid fails, but they require appropriate fuel or energy supply, testing, maintenance, and switching systems. A backup component is useful only if it is sized correctly, maintained, and able to start and connect when needed.

8.3 Redundant power paths

Some facilities use multiple power paths and equipment designed so maintenance or a component failure does not interrupt service. Redundancy levels are often described using terms such as N, N+1, or 2N. In simplified terms, N is the capacity required to support the load; N+1 adds one extra component or capacity unit; 2N provides two complete capacity paths. Real designs differ, and labels alone do not prove operational performance.

8.4 Common-mode failures

Two backup systems can fail together if they share a control system, room, fuel supply, cooling system, or human procedure. This is called a common-mode risk. Reliability engineering therefore examines dependencies, physical separation, maintenance windows, testing, and the ability to operate safely during unusual conditions.

> *Important distinction: backup power reduces some risks; it does not guarantee uninterrupted service. A facility can remain powered while an application is unavailable, or an application can fail because of software even when every physical system is healthy.*

9. Cooling and thermal management

9.1 Why heat must be removed

Processors, memory, storage, and power electronics convert electrical energy into heat during operation. Excessive temperature can cause components to reduce performance, shut down, or fail. Cooling systems transfer heat away from equipment and release it to the surrounding environment through mechanical or other heat-rejection systems.

9.2 Air cooling

Many facilities use fans to move air through server equipment and cooling systems to condition the air entering the data hall. Good airflow design separates hot and cold air where practical, avoids obstructions, and uses sensors to detect unusual conditions. Poor airflow can create local hot spots even when the room's average temperature appears acceptable.

9.3 Liquid cooling

Liquid cooling moves heat through a fluid and can be useful for equipment with high heat density. Designs include heat exchangers and direct-to-chip systems; immersion approaches place equipment in a suitable dielectric fluid. These designs involve additional engineering, leak management, maintenance, and compatibility considerations. They are not automatically better in every location or for every workload.

9.4 Trade-offs

Cooling choices involve electricity, water, climate, reliability, cost, equipment density, and local environmental conditions. Evaporative cooling can reduce some electricity requirements in suitable conditions but consumes water. Dry cooling can reduce on-site water consumption but may use more electricity under some conditions. The best choice is site-specific.

9.5 Monitoring and response

Operators monitor inlet temperatures, humidity, cooling capacity, pumps, fans, and alarms. If cooling capacity is lost, response may include shifting workloads, reducing load, switching equipment, or safely shutting down affected systems. The correct response depends on the facility's design and the severity of the event.

10. Reliability, redundancy, and failure

10.1 Reliability is a system property

Reliability does not come from a single expensive server. It depends on the complete system: hardware, software, power, cooling, networks, data copies, staff, procedures, and external dependencies. A robust component can be undermined by a weak shared dependency.

10.2 Failure domains

A failure domain is a part of a system that can be affected by one event. It may be a server, rack, room, building, availability zone, region, or external provider. Spreading components across failure domains can reduce the chance that one event affects all copies, although separation adds cost and complexity.

10.3 Monitoring and incident response

Monitoring systems collect signals such as error rates, latency, temperatures, power status, and storage health. Alerts help teams detect abnormal conditions. During an incident, teams need to establish what changed, limit impact, restore service, communicate clearly, and preserve evidence for later analysis. Poorly tuned alerts can create noise, while missing telemetry can delay diagnosis.

10.4 Maintenance without unnecessary downtime

Equipment needs firmware updates, repairs, inspections, and replacement. Redundancy can allow maintenance while services continue, but only if the remaining capacity can handle the load and procedures are followed. Maintenance itself can create risk, especially if a second fault occurs while a redundant component is unavailable.

10.5 What happens when a component fails?

• A server fails: a load balancer or orchestration system may stop sending it new work.

• A drive fails: a storage system may reconstruct data from redundant copies, depending on design.

• A network link fails: routing or link aggregation may move traffic to an alternate path.

• A cooling unit fails: another unit may take over, or operators may reduce load.

• A whole region fails: applications designed for multi-region operation may continue elsewhere, but data consistency and dependencies can complicate recovery.

These are possible responses, not guarantees. Actual outcomes depend on architecture, configuration, capacity, and the nature of the failure.

11. Where data centres are located

11.1 Connectivity and users

Facilities may be located near major network connections, population centres, enterprise customers, or other infrastructure. Low latency matters for interactive services, while some batch workloads can tolerate longer delays. Network access and the cost of connecting a site can therefore influence location.

11.2 Electricity supply

Large facilities need sufficient electrical capacity and a dependable route to the grid. Availability, price, transmission capacity, and the emissions intensity of electricity can all matter. A claim that a facility uses “renewable energy” should be read with its definition, time period, geographic scope, and accounting method in mind.

11.3 Climate and water

Climate affects cooling options and operating costs. Water availability and local water stress can influence whether evaporative systems are suitable. A design that performs well in one region may be inappropriate in another. Local environmental conditions should be evaluated rather than assumed from a global average.

11.4 Land, planning, and community concerns

Data centres need land, grid connections, roads, fibre routes, and permits. Communities may consider jobs, tax revenue, land use, water demand, noise, and the opportunity cost of electrical capacity. These impacts vary by project and locality. A balanced case study should use local records and avoid assuming that every facility has the same effects.

11.5 Distributed and centralised designs

Centralised facilities can benefit from scale and specialised operations. Distributed facilities may reduce latency or provide local processing. The best arrangement depends on service requirements, cost, regulation, data locality, reliability, and network conditions. There is no single arrangement that is ideal for all workloads.

12. Security and physical access

12.1 Layers of security

Data-centre security commonly combines perimeter controls, controlled entry, identity checks, monitoring, visitor procedures, and restrictions on access to equipment areas. The exact controls vary with the operator, the sensitivity of the data, and legal or contractual requirements.

12.2 Physical and digital security overlap

A secure building does not make software automatically secure. Digital security also requires identity and access management, encryption, patching, vulnerability management, network segmentation, logging, and incident response. Conversely, strong software controls can be undermined if unauthorised people can access equipment or storage media.

12.3 Secure equipment retirement

When a server or storage device is retired, operators must manage data-bearing media. Processes may include secure erasure, reuse, or physical destruction, according to policy and the sensitivity of the information. Documentation and chain of custody help establish that the process was completed correctly.

12.4 Privacy and shared services

Cloud customers share a provider's physical infrastructure with other workloads in many service models. Isolation mechanisms and access controls are designed to prevent one customer from accessing another's data. Customers still need to understand the provider's responsibilities, their own configuration duties, and the relevant contract and compliance requirements.

13. People who operate the facility

Data centres are maintained by teams, not by machines alone. Data-centre technicians install equipment, replace parts, trace cables, and carry out checks. Facilities engineers oversee power and cooling systems. Network engineers manage connectivity and routing. Site reliability and software operations teams monitor service health and respond to incidents. Security staff manage physical access, while procurement and logistics teams help ensure that parts and supplies are available.

The work includes planned maintenance and urgent response. A fault can require coordination between teams because the visible symptom may not reveal the underlying cause. For example, a network problem could arise from a failed switch, a power issue, a software change, or a remote provider. Clear documentation, escalation paths, training, and practice exercises improve the chance of a safe and rapid response.

The human dimension is important to the website's central argument: online services depend on skilled work, procedures, and decisions. A responsible explanation should not imply that infrastructure runs itself or that automation removes the need for people.

14. Documented case study: the 2017 Google Cloud incident

A useful documented example is the Google Cloud incident on 11 March 2017, when an error in the handling of network capacity affected access to multiple Google Cloud services. Google's public incident report described how a network event propagated beyond the initially affected component and disrupted services. The case illustrates that cloud availability depends on the interaction of network architecture, capacity, configuration, and operational response.

The teaching value of this incident is not that every outage has the same cause. Rather, it shows why a large service can be affected by a change in one part of the infrastructure and why redundancy must be tested against realistic failure scenarios. A spare path or component is not sufficient if a shared dependency or configuration error affects it too.

What students should learn from the case

• A cloud service is a system of connected components, not one server.

• Network capacity and configuration can affect multiple services.

• The public explanation of an incident should be tied to the provider's report and date.

• An outage is evidence of a failure mode, not proof that all cloud services are unreliable.

Source check: use Google's original post-incident report for exact timing, affected services, root cause, and recovery details before publishing a detailed timeline. This report summarises the lesson and does not invent exact outage-duration or user-impact figures.

15. How to explain the topic interactively

15.1 Feature concept: Explore Inside a Data Centre

Create a labelled illustration with selectable components: server rack, storage, network switch, router, power distribution, UPS, generator, cooling system, sensors, and security entrance. When a visitor selects a component, show three concise items: what it does, why it matters, and what could happen if it fails.

15.2 Example interaction text

• **Server:** Processes application instructions. If unavailable, requests may be moved to another server if the service is designed for that.

• **Storage:** Keeps files and records. Redundant copies and tested backups help protect availability and recovery.

• **Switch:** Connects equipment in a network. Alternate links may help when a link or device fails.

• **UPS:** Bridges some power interruptions. It needs correct sizing, maintenance, and a safe transition plan.

• **Cooling:** Removes heat. Cooling failure may require a rapid response to protect equipment.

15.3 Failure scenario

Ask visitors to choose what happens if one cooling unit fails. Options might include: (A) the entire internet stops, (B) another unit may take over if the system has capacity, or (C) the data centre needs no cooling. Explain that B is the best general answer, but only under the stated assumption that the backup system is available and appropriately designed.

15.4 Reliability scenario

Present two fictional designs. Design A has one network connection and one power path. Design B has alternate network paths and redundant power equipment. Ask which is likely to tolerate more individual failures, then ask what shared dependencies could still affect Design B. The activity should show trade-offs in cost, complexity, and resilience rather than implying that redundancy guarantees perfection.

15.5 Accessibility and accuracy

Use readable labels, keyboard-accessible controls, alt text, and a text alternative to the diagram. Label the illustration as a simplified educational model. Do not claim it shows a real company's confidential layout or a visitor's actual traffic.

16. Key findings, limitations, and conclusion

16.1 Key findings

• Cloud services depend on physical data centres and network infrastructure.

• Servers process requests; storage systems preserve information; switches and routers move data.

• Power, cooling, security, and monitoring are essential parts of the facility.

• Redundancy can improve resilience but adds cost and cannot eliminate every failure.

• Data copies, replication, and backups serve related but different purposes.

• Location decisions involve connectivity, electricity, climate, water, land, and community considerations.

• People and procedures remain essential for maintenance, security, and incident response.

• Any real-world outage should be described using a direct, dated source and a clearly bounded claim.

16.2 Limitations

This report describes common designs rather than inspecting a specific data centre. Providers use different architectures and may not disclose details of their facilities. The incident case study is included as a documented learning example, but exact details should be taken from the original provider postmortem. No claims are made here about the current location of a reader's data, the performance of a particular provider, or a universal uptime percentage.

16.3 Conclusion

The cloud is best understood as a networked service built on real equipment and managed systems. Servers and storage provide computation and persistence; networks connect them; power and cooling keep them operating; security and reliability practices protect them; and skilled people maintain the whole system. Once visitors understand these layers, they can see why online services are powerful, but never entirely detached from the physical world.

17. Glossary

Availability: The extent to which a service is usable when needed.

Backup: A separate recovery copy used to restore data after loss or damage.

Bandwidth: The capacity of a connection to carry data over time.

Cache: A stored copy of data kept to speed up later access.

Cloud computing: Providing computing resources through a network as a service.

Data centre: A facility housing computing equipment and its supporting infrastructure.

Failover: Switching service to an alternate component or path after a failure.

Fibre-optic cable: A cable that carries information as light through optical fibre.

IaaS: Infrastructure as a Service; rented compute, storage, and network resources.

Latency: Time delay in data transfer or processing.

Load balancer: A system that distributes requests among available servers.

PaaS: Platform as a Service; a managed environment for deploying applications.

Redundancy: Extra or alternate capacity intended to reduce the impact of failure.

Replication: Maintaining copies of data or services across devices or systems.

Router: A device that forwards packets between IP networks.

SaaS: Software as a Service; a complete application delivered to users.

Server: A computer that provides resources or services to other systems.

Storage array: A coordinated collection of storage devices managed as a system.

Switch: A device that forwards traffic among devices on a local network.

UPS: Uninterruptible power supply; short-term power continuity equipment.

Virtualisation: Software abstraction that allows physical resources to host logical environments.

18. References and further reading

The following sources are appropriate starting points for verifying technical explanations and the case study. Use the original page or report, check its date and scope, and avoid quoting a statistic without its definition and measurement period.

1. **Google Cloud, “Google Cloud Incident #17007” / incident report (11 March 2017).**
   https://status.cloud.google.com/incident/cloud-networking/17007

2. **Google Cloud documentation, Overview of Google Cloud.**
   https://cloud.google.com/docs/overview

3. **National Institute of Standards and Technology (NIST), The NIST Definition of Cloud Computing, SP 800-145.**
   https://csrc.nist.gov/pubs/sp/800/145/final

4. **U.S. Department of Energy, data centre energy and efficiency resources.**
   https://www.energy.gov/eere/buildings/data-centers-and-servers

5. **Lawrence Berkeley National Laboratory, data-centre energy research and publications.**
   https://eta.lbl.gov/publications/united-states-data-center-energy

6. **Uptime Institute, Global Data Center Survey and infrastructure reliability research.**
   https://uptimeinstitute.com/resources/research-and-reports

7. **Internet Society, Internet infrastructure and network resilience resources.**
   https://www.internetsociety.org/issues/

8. **Cloudflare Learning Center, explanations of DNS, routing, caching, and content delivery networks.**
   https://www.cloudflare.com/learning/

1. **Google Cloud, “Google Cloud Incident #17007” / incident report (11 March 2017).**
   https://status.cloud.google.com/incident/cloud-networking/17007

2. **Google Cloud documentation, Overview of Google Cloud.**
   https://cloud.google.com/docs/overview

3. **National Institute of Standards and Technology (NIST), The NIST Definition of Cloud Computing, SP 800-145.**
   https://csrc.nist.gov/pubs/sp/800/145/final

4. **U.S. Department of Energy, data centre energy and efficiency resources.**
   https://www.energy.gov/eere/buildings/data-centers-and-servers

5. **Lawrence Berkeley National Laboratory, data-centre energy research and publications.**
   https://eta.lbl.gov/publications/united-states-data-center-energy

6. **Uptime Institute, Global Data Center Survey and infrastructure reliability research.**
   https://uptimeinstitute.com/resources/research-and-reports

7. **Internet Society, Internet infrastructure and network resilience resources.**
   https://www.internetsociety.org/issues/

8. **Cloudflare Learning Center, explanations of DNS, routing, caching, and content delivery networks.**
   https://www.cloudflare.com/learning/

9. **HM216, Behind the Screen — Group Briefing Report, Subtopic 2: Inside the Cloud: Data Centres and Physical Infrastructure.**